

ME599 Final Project

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1. Abstract

The project is about dynamics simulation of spring-powered jumping robot at presence of dynamics friction. Three exercises were done. To verify my coding technic, in Exercise 1, I used Newtonian dynamics to simulate an inclining bar under dynamics friction, whose simulation result was verified by physical intuition. The switching before dynamics friction and static friction is modeled with a continuous function, and this modeling technic turn out to significantly reduce computation cost. In Exercise 2, the dynamics of jumping rail is formulated with Lagrangian dynamics. The simulation result is first verified by “degrading” the jumping rail down to an inclined bar, and then further verified with physical intuitions. Some design insights are produced out of the simulation result. In Exercise 3, I succumbed to the complicated algebra and failed to model the dynamics of jumping linkage.

2. Introduction

2.1 Motivation

In this project, I studied and exercised dynamics simulation of jumping at presence of general Coulomb friction. The project is partially motivated by my 2019 jumping vehicle project, where I tried to understand why the vehicle would pitch down after takeoff (Figure 2.1). The project is also motivated by my futural plan. I plan to design a jumping linkage to help my ornithopter take off (Figure 2.2). This design would require a co-simulation of a customized aerodynamics modeling with a multi-body dynamics simulation.



Figure 2.1 Earlier jumping rail experiment in 2019. The jumping vehicle pitch down (rotate such that its head is pointing downward) significantly after takeoff.

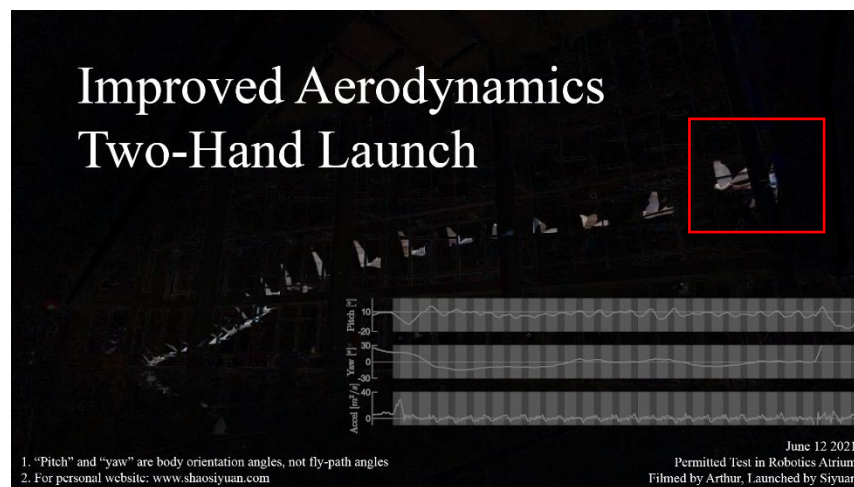


Figure 2.2 Look at the red rectangle, you will find that I throw the ornithopter into the sky by hands. Ideally, it should one day take-off by jumping.

2.2 Why Writing Code

Why not commercial code? There are commercial codes such as MSC ADAM that does dynamics simulation very well. However, one may still want to write a customized dynamics code for following purpose. First, by writing a customized code, I can decipher the black box of a commercial code, so that I have a better understanding of dynamics simulation. Second, since I will have all intermediate data of my customized code, I can better understand the simulation result and distinguish between the valuable physical phenomena and the “fake” phenomena induced by numerical issue. Third, a customized code can easily be integrated into an optimization code, which will help produce design insights.

3. Literature Review

Researchers at Berkley developed a high-performance jumping robot [1]. They used kinematics design optimization code to create ideal linkage [2]. Then, they specialized spring constant through dynamics simulation for vertical taking-off, where the friction is mostly static friction. According to what I knew, they did not simulate any inclined taking-off where dynamics friction would play a role. Indeed, there are some researchers that used Lagrangian dynamics with constraint to solve the jumping linkage dynamics [3]. I tried to taught myself Lagrangian dynamics with constraints (combining Lagrangian dynamics with Lagrangian multiplier) but failed. Thus, this project was done with Newtonian dynamics and Lagrangian dynamics

4. Methodology

I broke the project down to three exercises. Exercise 1 would be an inclined bar that fall onto the table at presence of dynamic friction (Figure 4.1). The simulation result can be easily verified with physical intuition. If verified, I would move on to Exercise 2, where I constrain a point mass onto the rail (the bar in previous exercise), and then added a spring between the point mass and the rail (Figure 4.2). Once released, the spring would introduce jumping motion. Exercise 2 would be verified by “degrading” it down to Exercise 1, which would be achieved through fixing the point mass onto the center of mass of the rail. Then, I would move to Exercise 3, where I replace the rail with a linkage (Figure 4.3). Exercise 3 failed.

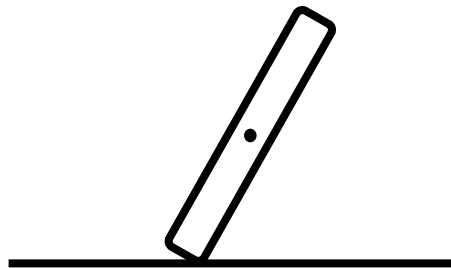


Figure 4.1 Inclined bar

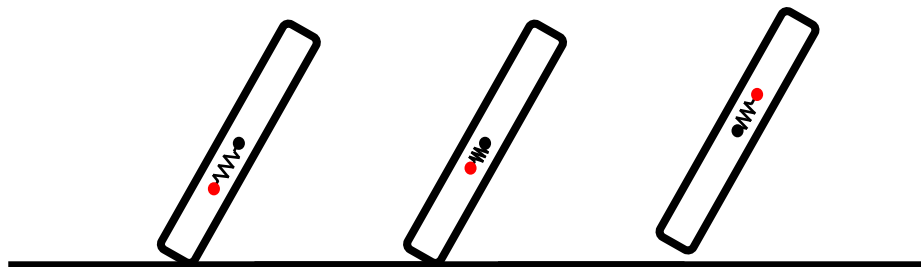


Figure 4.2 Jumping rail

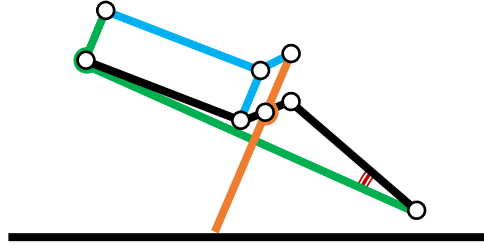
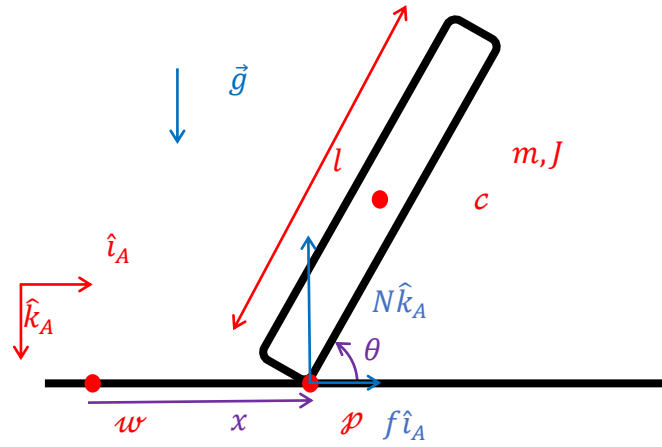


Figure 4.3 Jumping linkage with torsional spring.

5. Exercise 1

5.1 Formulation

A bar with center of mass c , length l , and inertia m, J is sliding and rotating. w is a reference point with zero inertia acceleration, and $\{\hat{i}_A, \hat{j}_A, \hat{k}_A\}$ are an inertia frame satisfying right-hand rule. p is the point where the bar contacts the ground. The only nonconservative force is the friction $f\hat{i}_A$, note that f can take negative values. The two generalized coordinates are x and θ



The Newtonian dynamics formulation is as follows (eq. 5.1, 5.2)

$$J\ddot{\theta}\hat{j}_A = \sum \vec{M}_{B/c} = \left(\frac{1}{2}fl \sin \theta - \frac{1}{2}Nl \cos \theta \right) \hat{j}_A \quad (\text{eq. 5.1})$$

$$\left(\ddot{x} - \ddot{\theta} \frac{l}{2} \sin \theta - \dot{\theta}^2 \frac{l}{2} \cos \theta \right) \hat{i}_A + \left(-\ddot{\theta} \frac{l}{2} \cos \theta + \dot{\theta}^2 \frac{l}{2} \sin \theta \right) \hat{k}_A = {}^A\ddot{\vec{r}}_{c/w}$$

$${}^A\ddot{\vec{r}}_{c/w} = \frac{1}{m} \sum \vec{F} = \frac{f}{m} \hat{i}_A + \left(g - \frac{N}{m} \right) \hat{k}_A \quad (\text{eq. 5.2})$$

I used following simplification to model both the static friction and dynamics friction. Note that the $s(\dot{x})$ always takes value within $[-1, 1]$.

$$f = -s(\dot{x})\mu N, \quad s(\dot{x}) = \begin{cases} \text{sgn } \dot{x} & |\dot{x}| > \varepsilon \\ \dot{x}/\varepsilon & |\dot{x}| \leq \varepsilon \end{cases} \quad (\text{eq. 5.3})$$

Plugging (eq. 5.3) into (eq. 5.1) and (eq. 5.2), I managed to evolve $(\ddot{x}, \ddot{\theta}, N, f)$ of step $k + 1$ by doing some calculation about the $(x, \dot{x}, \theta, \dot{\theta})$ of step k (eq. 5.4).

$$\begin{bmatrix} \ddot{x} \\ \ddot{\theta} \\ N \\ f \end{bmatrix} = \text{inv} \left(\begin{bmatrix} 0 & \frac{1}{2}l \cos \theta & -\frac{1}{m} & 0 \\ 1 & -\frac{1}{2}l \sin \theta & 0 & -\frac{1}{m} \\ 0 & J & \frac{1}{2}l \cos \theta & -\frac{1}{2}l \sin \theta \\ 0 & 0 & s(\dot{x})\mu & 1 \end{bmatrix} \right) \begin{bmatrix} \dot{\theta}^2 \frac{1}{2}l \sin \theta - g \\ \dot{\theta}^2 \frac{1}{2}l \cos \theta \\ 0 \\ 0 \end{bmatrix} \quad (\text{eq. 5.4})$$

By further rearranging (eq. 5.4), I get equation (eq. 5.5), which I used MATLAB ode45 get numerical simulation.

$$\dot{q} = f(t, q), \quad q \triangleq \begin{bmatrix} x \\ \dot{x} \\ \theta \\ \dot{\theta} \end{bmatrix} \quad (\text{eq. 5.5})$$

5.2 Experiment

I give parameters following numerical values $m = 1\text{kg}$, $g = 9.8\text{N/kg}$, $l = 1\text{m}$, $J = ml^2/12$. First, let me simulate the frictionless case $\mu = 0$. Since there is no net force in horizontal direction, we expect the center of mass to be moving purely vertically, and the simulation model agree with my expectation. (Figure 5.1)

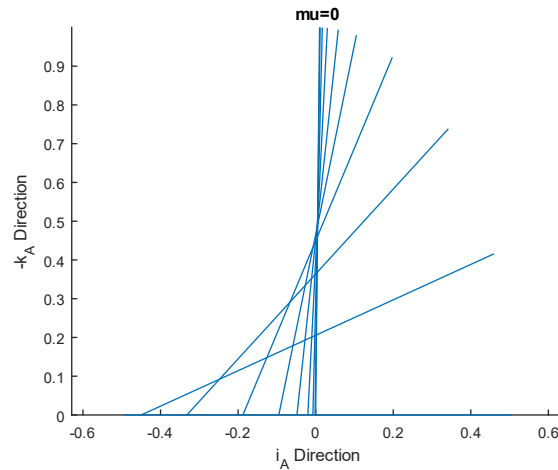


Figure 5.1 $x(0) = 0$, $\dot{x}(0) = 0$, $\theta(0) = \pi/2 - 0.01$, $\dot{\theta}(0) = 0$

Then, under same initial condition, I increased μ to 0.3. The result seems to make sense (Figure 5.2).

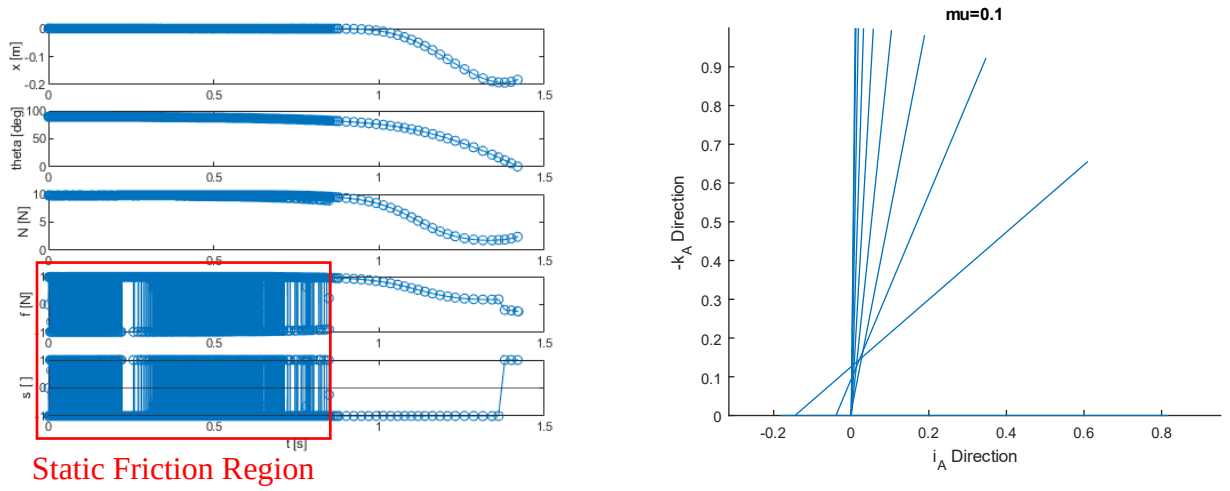


Figure 5.2 $\mu = 0.1$, $x(0) = 0$, $\dot{x}(0) = 0$, $\theta(0) = \pi/2 - 0.01$, $\dot{\theta}(0) = 0$

However, this 1.5s simulation takes the ode45 solver 16953 steps to solve, and all those calculations happen within the static friction region.

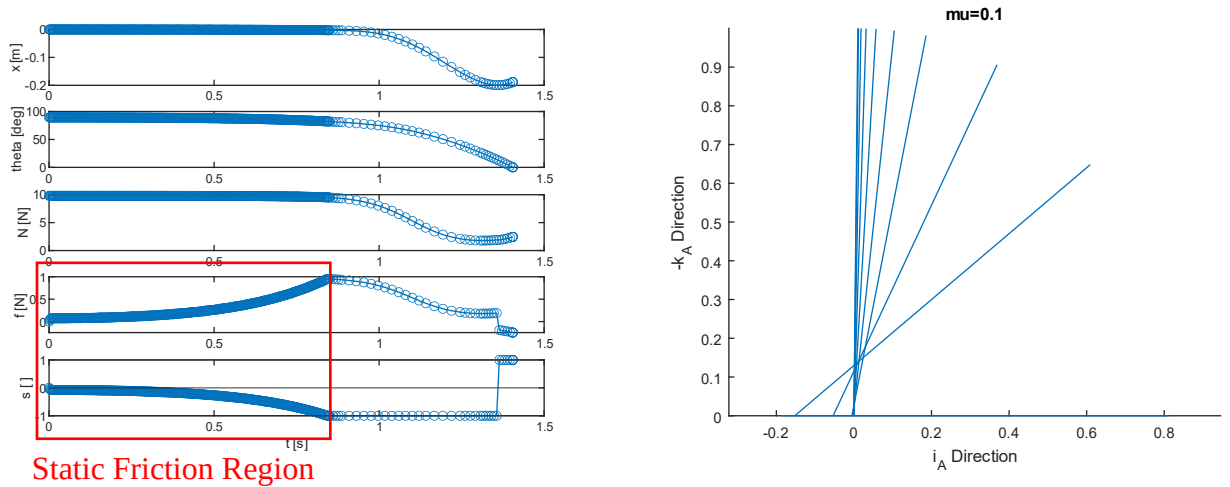


Figure 5.3 $\mu = 0.1$, $x(0) = 0$, $\dot{x}(0) = 0$, $\theta(0) = \pi/2 - 0.01$, $\dot{\theta}(0) = 0$ with $\epsilon = 1 \text{ cm/s}$

To save those unnecessary computations, I tuned the ϵ in (eq. 5.3) from 10^{-6} m/s to a more generous 10^{-2} m/s . This is to say, I consider any sliding velocity smaller than 1 cm/s as nonslip. This tuning significantly enhanced computational efficiency, and now the simulation only takes the ode45 solver 441 steps to solve (Figure 5.3).

Let me also check the simulation result for other initial conditions (Figure 5.4, 5.5)

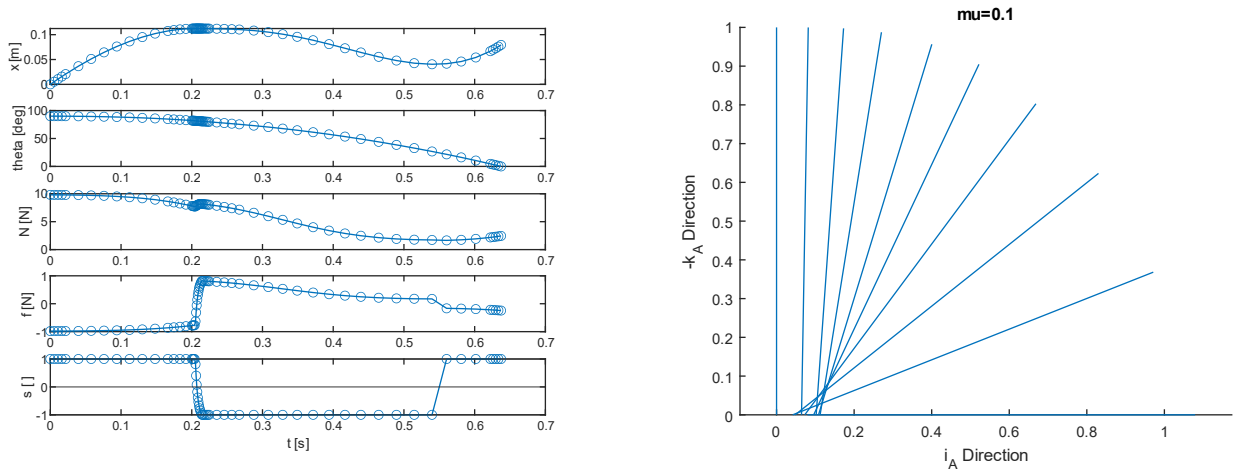


Figure 5.4 $\mu = 0.1$, $x(0) = 0$, $\dot{x}(0) = 1$, $\theta(0) = \pi/2$, $\dot{\theta}(0) = 0$

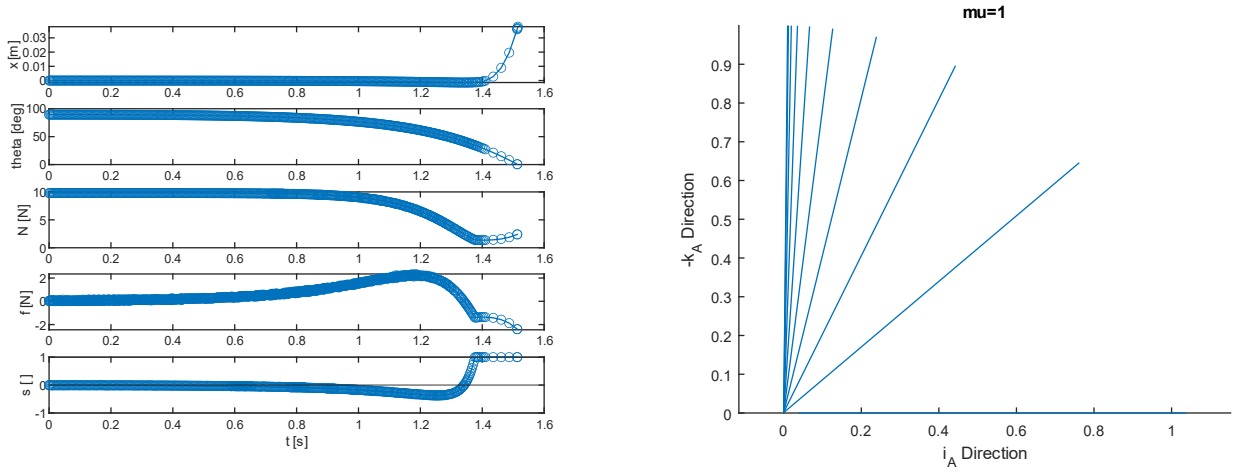


Figure 5.5 $\mu = 1$, $x(0) = 0$, $\dot{x}(0) = 0$, $\theta(0) = \pi/2 - 0.01$, $\dot{\theta}(0) = 0$

They all make senses to me. Thus, I move on to Exercise 2.

6. Exercise 2

6.1 Formulation

Now, let me replace the bar in Exercise 1 with a linear rail, with length l , mass m , center of mass c , and moment of inertia J . Let me put a particle d with mass M into the linear rail. This means, $\vec{r}_{c/d}$ is always parallel to $\cos \theta \hat{i}_A - \sin \theta \hat{k}_A$ (Figure 6.1).

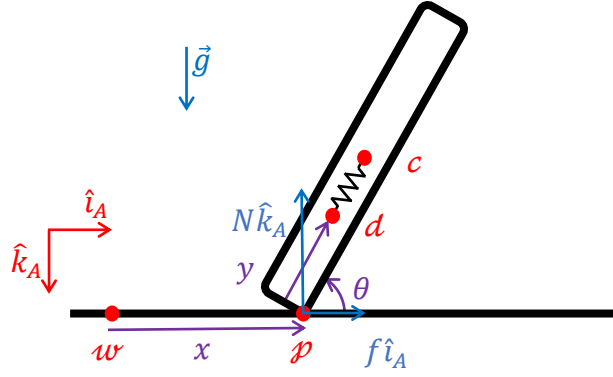


Figure 6.1 Schematic for Jumping Rail

Let me use a spring to connect the particle d to the center of mass of the rail c , and set the free length of the spring to be 0. Note that the elastic energy of the spring is a function of y , and as $E_k(y)$. If the spring has constant stiffness k , then $E_k(y)$ would be specialized to $k(l/2 - y)^2/2$. However, for reasons that will be explained later, I want to model the spring with a general non-constant stiffness, and keep the elastic energy $E_k(y)$ in its general form.

I transit to Lagrangian dynamics, so as to prepare myself for my later linkage dynamics formulation. I focused on the takeoff stage, where contact force $N \geq 0$ and p stays on the ground. Thus, I have three generalized coordinates, with which I represented the displacement vectors of c, d, p (eq. 6.1)

$$q \triangleq \begin{bmatrix} x \\ y \\ \theta \end{bmatrix}, \quad \vec{r}_{d/w} = \begin{bmatrix} x \\ 0 \\ 0 \end{bmatrix}^T \begin{bmatrix} \hat{i}_A \\ \hat{j}_A \\ \hat{k}_A \end{bmatrix}, \quad \vec{r}_{d/w} = \begin{bmatrix} x + y \cos \theta \\ 0 \\ -y \sin \theta \end{bmatrix}^T \begin{bmatrix} \hat{i}_A \\ \hat{j}_A \\ \hat{k}_A \end{bmatrix}, \quad \vec{r}_{c/w} = \begin{bmatrix} x + \frac{1}{2}l \cos \theta \\ 0 \\ -\frac{1}{2}l \sin \theta \end{bmatrix}^T \begin{bmatrix} \hat{i}_A \\ \hat{j}_A \\ \hat{k}_A \end{bmatrix} \quad (\text{eq. 6.1})$$

I further get velocity

$$\overset{A}{\vec{r}}_{p/w} = \begin{bmatrix} \dot{x} \\ 0 \\ 0 \end{bmatrix}^T \begin{bmatrix} \hat{i}_A \\ \hat{j}_A \\ \hat{k}_A \end{bmatrix}, \quad \overset{A}{\vec{r}}_{d/w} = \begin{bmatrix} \dot{x} + \dot{y} \cos \theta - y \sin \theta \dot{\theta} \\ 0 \\ -\dot{y} \sin \theta - y \cos \theta \dot{\theta} \end{bmatrix}^T \begin{bmatrix} \hat{i}_A \\ \hat{j}_A \\ \hat{k}_A \end{bmatrix}, \quad \overset{A}{\vec{r}}_{c/w} = \begin{bmatrix} \dot{x} - \frac{1}{2}l \sin \theta \dot{\theta} \\ 0 \\ -\frac{1}{2}l \cos \theta \dot{\theta} \end{bmatrix}^T \begin{bmatrix} \hat{i}_A \\ \hat{j}_A \\ \hat{k}_A \end{bmatrix}$$

This gives the kinetic energy T (eq. 6.2)

$$T = T_{\text{bar}} + T_d$$

$$T_d = \frac{1}{2}M \left\| \overset{A}{\vec{r}}_{d/w} \right\|^2$$

$$\begin{aligned}
&= \frac{1}{2}M(\dot{x}^2 + 2\dot{x}(\dot{y}\cos\theta - y\sin\theta\dot{\theta}) + \dot{y}^2 + \dot{\theta}^2y^2) \\
&= \dot{x}^2\left(\frac{1}{2}M\right) + \dot{x}\dot{y}(M\cos\theta) + \dot{x}\dot{\theta}(-My\sin\theta) + \dot{y}^2\left(\frac{1}{2}M\right) + \dot{\theta}^2\left(\frac{1}{2}My^2\right) \\
T_{\text{bar}} &= \frac{1}{2}m\left\|\vec{r}_{c/w}^{A\cdot}\right\|^2 + \frac{1}{2}J\dot{\theta}^2 \\
&= \frac{1}{2}m\left(\dot{x}^2 + \dot{x}(\dot{y}\cos\theta - y\sin\theta\dot{\theta}) + \frac{1}{4}\dot{y}^2 + \frac{1}{4}\dot{\theta}^2y^2\right) + \frac{1}{2}J\dot{\theta}^2 \\
&= \dot{x}^2\left(\frac{1}{2}m\right) + \dot{x}\dot{\theta}\left(-\frac{1}{2}ml\sin\theta\right) + \dot{\theta}^2\left(\frac{1}{8}ml^2 + \frac{1}{2}J\right)
\end{aligned}$$

Taking partial derivatives with T_d , we get (eq. 6.2)

$$\frac{\partial}{\partial t} \frac{\partial T_d}{\partial \dot{x}} = \ddot{x}(M) + \ddot{y}(M\cos\theta) + \dot{y}\dot{\theta}(-M\sin\theta) + \ddot{\theta}(-My\sin\theta) + \dot{\theta}^2(-My\cos\theta)$$

$$\frac{\partial T_d}{\partial x} = 0$$

$$\frac{\partial}{\partial t} \frac{\partial T_d}{\partial \dot{y}} = \ddot{x}(M\cos\theta) + \dot{x}\dot{\theta}(-M\sin\theta) + \ddot{y}(M)$$

$$\frac{\partial T_d}{\partial y} = \dot{x}\dot{\theta}(-M\sin\theta) + \dot{\theta}^2(My)$$

$$\frac{\partial}{\partial t} \frac{\partial T_d}{\partial \dot{\theta}} = \ddot{x}(-My\sin\theta) + \dot{x}\dot{y}(-M\sin\theta) + \dot{x}\dot{\theta}(-My\cos\theta) + \ddot{\theta}(My^2) + \dot{y}\dot{\theta}(2My)$$

$$\frac{\partial T_d}{\partial \theta} = \dot{x}\dot{y}(-M\sin\theta) + \dot{x}\dot{\theta}(-My\cos\theta)$$

By rearranging them, we get (eq. 6.3)

$$\left\{ \begin{aligned} \frac{\partial}{\partial t} \frac{\partial T_d}{\partial \dot{x}} - \frac{\partial T_d}{\partial x} &= \ddot{x}(M) + \ddot{y}(M\cos\theta) + \ddot{\theta}(-My\sin\theta) + \dot{y}\dot{\theta}(-M\sin\theta) + \dot{\theta}^2(-My\cos\theta) \\ \frac{\partial}{\partial t} \frac{\partial T_d}{\partial \dot{y}} - \frac{\partial T_d}{\partial y} &= \ddot{x}(M\cos\theta) + \ddot{y}(M) + \dot{\theta}^2(-My) \\ \frac{\partial}{\partial t} \frac{\partial T_d}{\partial \dot{\theta}} - \frac{\partial T_d}{\partial \theta} &= \ddot{x}(-My\sin\theta) + \ddot{\theta}(My^2) + \dot{y}\dot{\theta}(2My) \end{aligned} \right. \quad (\text{eq. 6.3})$$

By taking partial derivative about T_{bar} , we get (eq. 6.4)

$$\frac{\partial}{\partial t} \frac{\partial T_{\text{bar}}}{\partial \dot{x}} = \ddot{x}(m) + \ddot{\theta} \left(-\frac{1}{2} ml \sin \theta \right) + \dot{\theta}^2 \left(-\frac{1}{2} ml \cos \theta \right)$$

$$\frac{\partial T_{\text{bar}}}{\partial x} = 0$$

$$\frac{\partial}{\partial t} \frac{\partial T_{\text{bar}}}{\partial \dot{y}} = 0$$

$$\frac{\partial T_{\text{bar}}}{\partial y} = 0$$

$$\frac{\partial}{\partial t} \frac{\partial T_{\text{bar}}}{\partial \dot{\theta}} = \ddot{x} \left(-\frac{1}{2} ml \sin \theta \right) + \dot{x} \dot{\theta} \left(-\frac{1}{2} ml \cos \theta \right) + \ddot{\theta} \left(\frac{1}{4} ml^2 + J \right)$$

$$\frac{\partial T}{\partial \theta} = \dot{x} \dot{\theta} \left(-\frac{1}{2} ml \cos \theta \right)$$

$$\left\{ \begin{array}{l} \frac{\partial}{\partial t} \frac{\partial T_{\text{bar}}}{\partial \dot{x}} - \frac{\partial T_{\text{bar}}}{\partial x} = \ddot{x}(m) + \ddot{\theta} \left(-\frac{1}{2} ml \sin \theta \right) + \dot{\theta}^2 \left(-\frac{1}{2} ml \cos \theta \right) \\ \frac{\partial}{\partial t} \frac{\partial T_{\text{bar}}}{\partial \dot{y}} - \frac{\partial T_{\text{bar}}}{\partial y} = 0 \\ \frac{\partial}{\partial t} \frac{\partial T_{\text{bar}}}{\partial \dot{\theta}} - \frac{\partial T}{\partial \theta} = \ddot{x} \left(-\frac{1}{2} ml \sin \theta \right) + \dot{x} \dot{\theta} \left(-\frac{1}{2} ml \cos \theta \right) + \ddot{\theta} \left(\frac{1}{4} ml^2 + J \right) \end{array} \right. \quad (\text{eq. 6.4})$$

Then, let me deal with potential energy (eq. 6.5)

$$V = \frac{1}{2} mgl \sin \theta + Mgy \sin \theta + E_k(y)$$

$$\left\{ \begin{array}{l} \frac{\partial V}{\partial x} = 0 \\ \frac{\partial V}{\partial y} = Mgy \sin \theta + \frac{\partial E_k(y)}{\partial y} \\ \frac{\partial V}{\partial \theta} = \frac{1}{2} mgl \cos \theta + Mgy \cos \theta \end{array} \right. \quad (\text{eq. 6.5})$$

Then, let me deal with non-conservative external force f and transform it into generalized forces Q (eq. 6.6)

$$f = -s(\dot{x})\mu N, \quad s(\dot{x}) = \begin{cases} \text{sgn } \dot{x} & |\dot{x}| > \varepsilon \\ \dot{x}/\varepsilon & |\dot{x}| \leq \varepsilon \end{cases}$$

$$Q = f \hat{i}_A \cdot \begin{bmatrix} \frac{\partial \vec{r}_{p/w}}{\partial x} \\ \frac{\partial \vec{r}_{p/w}}{\partial y} \\ \frac{\partial \vec{r}_{p/w}}{\partial \theta} \end{bmatrix} = f \hat{i}_A \cdot \begin{bmatrix} \hat{i}_A \\ \vec{0} \\ \vec{0} \end{bmatrix} = \begin{bmatrix} f \\ 0 \\ 0 \end{bmatrix} \quad (\text{eq. 6.6})$$

Then, let me deal with the reaction force N

$$-N + (M + m)g = \hat{k}_A \cdot \left(M \vec{r}_{d/w}^{A..} + m \vec{r}_{c/w}^{A..} \right)$$

$$\hat{k}_A \cdot \vec{r}_{d/w}^{A..} = -\ddot{y} \sin \theta - 2\dot{y}\dot{\theta} \cos \theta - \ddot{\theta} y \cos \theta + \dot{\theta}^2 y \sin \theta$$

$$\hat{k}_A \cdot \vec{r}_{c/w}^{A..} = \dot{\theta}^2 \frac{1}{2} l \sin \theta + \ddot{\theta} \left(-\frac{1}{2} l \cos \theta \right)$$

Thus, we get the expression for reaction force N

$$N = g(M + m) + \ddot{y}(M \sin \theta) + \ddot{\theta} \left(\frac{1}{2} ml \cos \theta + My \cos \theta \right) + \dot{y}\dot{\theta}(2M \cos \theta) + \dot{\theta}^2 \left(-My \sin \theta - \frac{1}{2} ml \sin \theta \right)$$

Combining everything we had, we get the $\mathcal{A}\chi = \mathcal{B}$ form (eq. 6.7)

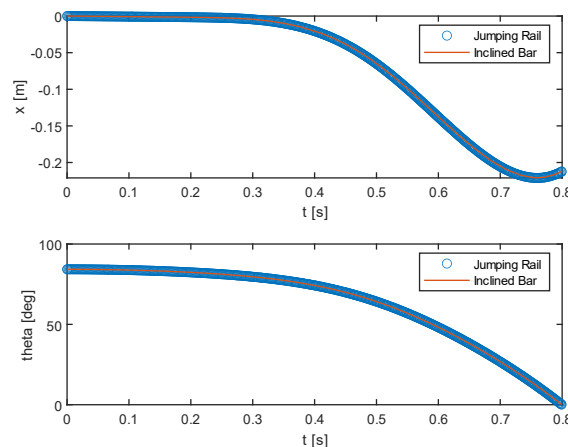
$$\mathcal{A} = \begin{bmatrix} M + m & M \cos \theta & -My \sin \theta - \frac{1}{2} ml \sin \theta & 0 & -1 \\ M \cos \theta & M & 0 & 0 & 0 \\ -My \sin \theta - \frac{1}{2} ml \sin \theta & 0 & My^2 + \frac{1}{4} ml^2 + J & 0 & 0 \\ 0 & M \sin \theta & \frac{1}{2} ml \cos \theta + My \cos \theta & -1 & 0 \\ 0 & 0 & 0 & s(\dot{x})\mu & 1 \end{bmatrix}, \quad \chi = \begin{bmatrix} \ddot{x} \\ \ddot{y} \\ \ddot{\theta} \\ N \\ f \end{bmatrix} \quad (\text{eq. 6.7})$$

$$\mathcal{B} = \begin{bmatrix} \dot{y}\dot{\theta}(M \sin \theta) + \dot{\theta}^2 \left(My \cos \theta + \frac{1}{2} ml \cos \theta \right) \\ \dot{\theta}^2(My) - Mg \sin \theta - \frac{\partial E_k(y)}{\partial y} \\ -\dot{y}\dot{\theta}(2My) - \frac{1}{2} mgl \cos \theta - Mgy \cos \theta \\ -g(M + m) - \dot{y}\dot{\theta}(2M \cos \theta) + \dot{\theta}^2 \left(My \sin \theta + \frac{1}{2} ml \sin \theta \right) \\ 0 \end{bmatrix}, \quad q = \begin{bmatrix} x \\ \dot{x} \\ y \\ \dot{y} \\ \theta \\ \dot{\theta} \end{bmatrix}$$

6.2 Experiment

The diagram shows a mechanical system. A thick black bar is tilted at an angle θ relative to a horizontal surface. A spring is attached to the bar at a distance d from the pivot point p . The spring is labeled c . A vertical arrow labeled $\vec{g}$ indicates the direction of gravity. A horizontal arrow labeled $\hat{i}_A$ and a vertical arrow labeled $\hat{k}_A$ are shown at point w . A horizontal arrow labeled x and a vertical arrow labeled y are shown at point p . A horizontal arrow labeled $f\hat{i}_A$ and a vertical arrow labeled $N\hat{k}_A$ are shown at point p . The pivot point is labeled p .

Indeed, when I set the stiffness to a huge value ($k = 10^5 N/m$), the inclined bar model and the jumping rail model give exactly the same prediction (Figure 6.3). The detailed initial conditions for the jumping rail model is $x(0) = 0, \dot{x}(0), \theta(0) = \pi/2 - 0.1, \dot{\theta}(0) = 0, y(0) = 0.5 - 9.8/10^5, \dot{y}(0) = 0$, and the detailed initial condition for the inclined bar model is $x(0) = 0, \dot{x}(0), \theta(0) = \pi/2 - 0.1, \dot{\theta}(0) = 0$.



Since the “degraded” case works well, then I change the spring stiffness to a realistic value ($k = 10^3 N/m$) continue to check the vertical jumping case with following parameters

```

para=struct();
para.g=9.8; %N/kg
para.M=0.4; %kg
para.m=0.1; %kg
para.mu=0.3; %N/N
para.l=0.5; %m
para.J=1/12*1*para.l^2;
k=1e2; %N/m

```

The result makes sense (Figure 6.4). As expected, there is negligible sliding motion or inclining motion ($x \approx 0, \dot{x} \approx 0, \dot{\theta} \approx 0$). There is no friction at all ($s = 0, f \approx 0$). What's more, when the rail is just about to take off, its weight should be supported purely by the elongated spring, and this agrees with the simulation result since $k(y_{\text{takeoff}} - y_0) = 10^2 \text{ N/kg} \times (0.260011 - 0.25) \text{ m} = 1.0011 \text{ N} \approx 0.98 \text{ N} = mg$. Thus, the result makes sense.

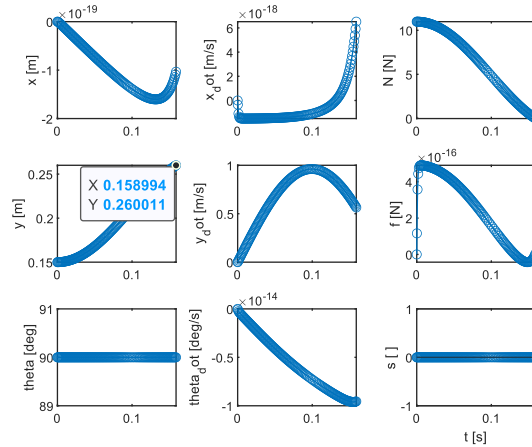


Figure 6.4 Simulation result with $x(0) = 0, \dot{x}(0), \theta(0) = \pi/2, \dot{\theta}(0) = 0, y(0) = l/2 - 0.1, \dot{y}(0) = 0$

Then, let me keep everything else unchanged and set $\theta(0) = \pi/4$. As expected, the simulation immediately figured out that the friction should be dynamics friction (look at how $s(\dot{x})$ function switched from 0 to the -1 in red block), and the contact point slides to the left (Figure 6.5).

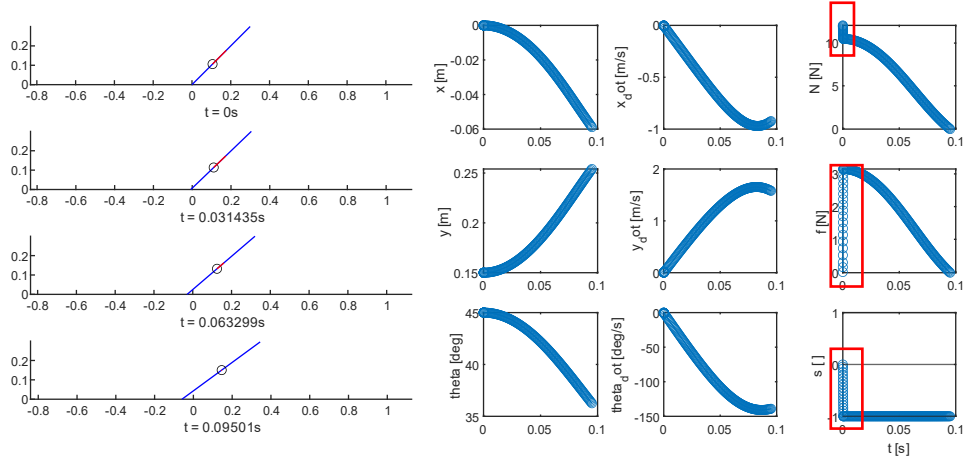


Figure 6.5 Simulation result with $x(0) = 0, \dot{x}(0), \theta(0) = \pi/4, \dot{\theta}(0) = 0, y(0) = l/2 - 0.1, \dot{y}(0) = 0$

6.3 Design Insights

Since the tool is verified, I move on to use the tool to generate some design insights. I care about 3 takeoff performance: the horizontal term of effective takeoff speed $\bar{v}_{\text{hor,takeoff}}$, and vertical term of effective takeoff speed $\bar{v}_{\text{vert,takeoff}}$, and the pitch rate $\dot{\theta}_{\text{takeoff}}$

$$\bar{v}_{\text{hor,takeoff}} \triangleq \hat{l}_A \cdot \frac{M \vec{r}_{d/w}^{A\cdot} + m \vec{r}_{c/w}^{A\cdot}}{M + m} \bigg|_{t=t_{\text{takeoff}}}, \quad \bar{v}_{\text{vert,takeoff}} \triangleq -\hat{k}_A \cdot \frac{M \vec{r}_{d/w}^{A\cdot} + m \vec{r}_{c/w}^{A\cdot}}{M + m} \bigg|_{t=t_{\text{takeoff}}},$$

$$\dot{\theta}_{\text{takeoff}} \triangleq \dot{\theta} \big|_{t=t_{\text{takeoff}}}$$

The result is as follows (Figure 6.6). The middle plot agrees with my expectation that more vertical initial condition ($\theta \rightarrow \pi/2$) would produce a larger vertical takeoff speed. The right-hand-side plot show that significant pitching motion will be observed during the jumping, which agree with the earlier experiment result of jumping rail (Figure 6.7).



Figure 6.7 Earlier jumping rail experiment in 2019.

The left-hand-side plot is the most interesting one. Within the static friction region (this region is characterized by $\theta_0 \geq \text{atan } 1/\mu$, and the $\theta_0 = \text{atan } 1/\mu$ line is plotted in red in the left-hand-side plot), the horizontal takeoff speed purely depends on θ_0 . For any given μ , to get maximum horizontal takeoff speed, we would need to enter the dynamic friction region and select the takeoff angle θ_0 around the blue curve ($\theta_0 = \text{atan } 1/\mu - 10^\circ$)

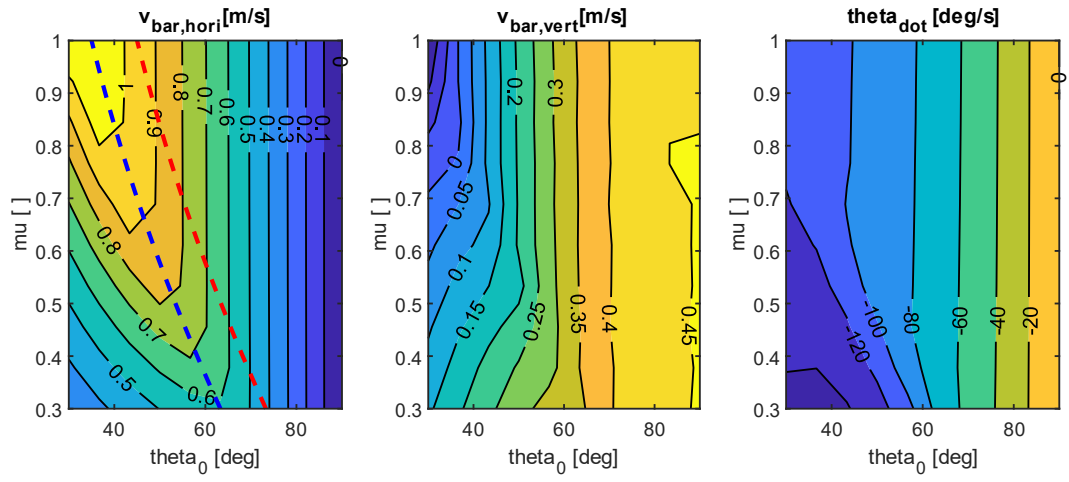


Figure 6.6 Contour plot result of simulation with the same linear spring $k = 10^2 \text{ N/kg}$ with an initial compression of 0.1 m

7. Exercise 3

7.1 Formulation

The black link and the green link form the famous Watt's linkage, where the pivot point d moves almost rectilinearly with respect to the green link (Figure 7.1). With the help of two pairs of concatenated parallelogram linkage (blue links), the orange link is constraint to be always perpendicular to the green link. To simplify the problem, I assume all links to be light weighted except the green link and the orange link. Meanwhile, assume the torsional spring on the angle $\angle b\sigma_4\sigma_2$ to be a linear spring with spring constant κ .

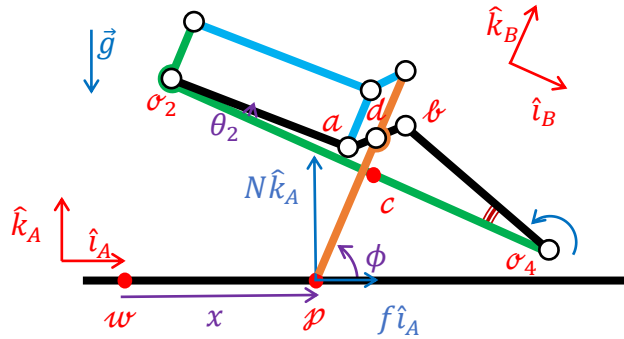


Figure 7.1 Linkage schematics

I have in total three generalized coordinates x, ϕ, θ_2 . When the green link is fixed on the ground (x, ϕ becomes constant), the kinematics is trivial (Figure 7.2), and all kinetic energy depends solely on θ_2 and its derivatives. However, when x, ϕ are not constants, the kinetic energy becomes very complicated. My algebra did not work out (Please see appendix). Exercise 3 failed.

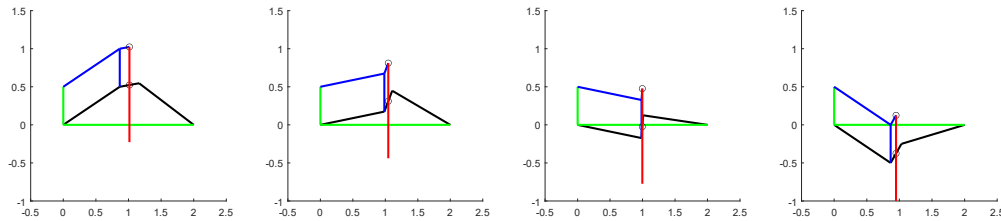


Figure 7.2 The kinematics of the linkage.

6. Discussion

The methodology indeed helped make the verification of each code much easier. This is good.

In the linkage dynamics part, I tried to use Lagrangian dynamics to formula the state equations. This process, however, involved lots of algebra: I spent lots of time deriving, add, subtracting, and differentiating those mathematical expression by hands, which was both painful and difficult.

Thus, the major problem is not dynamics, but algebra. There are two potential solutions. I can try figuring out a way to make algebra easier, and this may involve using computer to do symbolic calculation. I may also try using other formulation of dynamics, such as Lagrangian dynamics with constraint (Lagrangian multiplier), which can to some degree simplify the equation.

7. References

- [1] Haldane, Duncan W ; Plecnik, M M ; Yim, J K ; Fearing, R S, “Robotic vertical jumping agility via series-elastic power modulation,” Science robotics, vol. 1, no. 1. United States.
- [2] Plecnik, Mark M ; Haldane, Duncan W ; Yim, Justin K ; Fearing, Ronald S, “Design Exploration and Kinematic Tuning of a Power Modulating Jumping Monopod,” Journal of mechanisms and robotics, vol. 9, no. 1. ASME.
- [3] Long Bai, Wenjie Ge, Xiaohong Chen and Ruifeng Chen “Design and Dynamics Analysis of a Bio-Inspired Intermittent Hopping Robot for Planetary Surface Exploration” International Journal of Advanced Robotic Systems

Exercise 3: Failed Algebra

I have in total three generalized displacement x, ϕ, θ_2 . With the help of kinematics, I can get following relation (eq. 3.1), where θ_2 is the angle between green link and the $\sigma_2 a$ link, and θ_4 is the angle between green link and the $\sigma_4 b$ link.

$$\vec{r}_{d/\sigma_2} = \begin{bmatrix} \hat{i}_B \\ \hat{j}_B \\ \hat{k}_B \end{bmatrix}^T f(\theta_2), \quad {}^{B\cdot}\vec{r}_{d/\sigma_2} = \begin{bmatrix} \hat{i}_B \\ \hat{j}_B \\ \hat{k}_B \end{bmatrix}^T \left(\frac{\partial f}{\partial \theta_2}(\theta_2) \right) \dot{\theta}_2, \quad {}^{B\cdot\cdot}\vec{r}_{d/\sigma_2} = \begin{bmatrix} \hat{i}_B \\ \hat{j}_B \\ \hat{k}_B \end{bmatrix}^T \left(\left(\frac{\partial^2 f}{\partial \theta_2^2}(\theta_2) \right) \dot{\theta}_2^2 + \left(\frac{\partial f}{\partial \theta_2}(\theta_2) \right) \ddot{\theta}_2 \right)$$

Note that the orientation matrix between Frame A and Frame B are as follow

$$\begin{bmatrix} \hat{i}_B \\ \hat{j}_B \\ \hat{k}_B \end{bmatrix} = \mathcal{O}(\phi) \begin{bmatrix} \hat{i}_A \\ \hat{j}_A \\ \hat{k}_A \end{bmatrix}$$

$$\mathcal{O}(\phi) \triangleq \begin{bmatrix} \sin \phi & 0 & -\cos \phi \\ 0 & 1 & 0 \\ \cos \phi & 0 & \sin \phi \end{bmatrix}, \quad \dot{\mathcal{O}}(\phi) = \mathcal{O}\left(\phi + \frac{\pi}{2}\right)\dot{\phi}, \quad \ddot{\mathcal{O}}(\phi) = \mathcal{O}(\phi + \pi)\dot{\phi}^2 + \mathcal{O}\left(\phi + \frac{\pi}{2}\right)\ddot{\phi}$$

$$\vec{r}_{d/\sigma_2} = \begin{bmatrix} \hat{i}_B \\ \hat{j}_B \\ \hat{k}_B \end{bmatrix}^T f(\theta_2) = \begin{bmatrix} \hat{i}_A \\ \hat{j}_A \\ \hat{k}_A \end{bmatrix}^T [\mathcal{O}(\phi)]^T f(\theta_2) \Rightarrow \vec{r}_{d/\sigma_2}|_A = [\mathcal{O}(\phi)]^T f(\theta_2) = [\mathcal{O}(\phi)]^T \begin{bmatrix} f_i \\ 0 \\ f_k \end{bmatrix}$$

$$\left(\vec{r}_{d/\sigma_2}|_A \right)^\cdot = [\mathcal{O}(\phi + \frac{\pi}{2})]^T \begin{bmatrix} f_i' \\ 0 \\ f_k' \end{bmatrix} \dot{\phi} + [\mathcal{O}(\phi)]^T \begin{bmatrix} f_i' \\ 0 \\ f_k' \end{bmatrix} \dot{\theta}_2$$

$$\left(\vec{r}_{d/\sigma_2}|_A \right)^\cdot\cdot = [\mathcal{O}(\phi + \pi)]^T \begin{bmatrix} f_i \\ 0 \\ f_k \end{bmatrix} \dot{\phi}^2 + 2 [\mathcal{O}(\phi + \frac{\pi}{2})]^T \begin{bmatrix} f_i' \\ 0 \\ f_k' \end{bmatrix} \dot{\phi} \dot{\theta}_2 + [\mathcal{O}(\phi + \frac{\pi}{2})]^T \begin{bmatrix} f_i \\ 0 \\ f_k \end{bmatrix} \ddot{\phi} + [\mathcal{O}(\phi)]^T \begin{bmatrix} f_i'' \\ 0 \\ f_k'' \end{bmatrix} \dot{\theta}_2^2 + [\mathcal{O}(\phi)]^T \begin{bmatrix} f_i' \\ 0 \\ f_k' \end{bmatrix} \ddot{\theta}_2$$

$$\left(\vec{r}_{d/w}|_A \right)^\cdot = \left(\vec{r}_{p/w}|_A \right)^\cdot + \left(\vec{r}_{d/p}|_A \right)^\cdot = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \dot{x} + [\mathcal{O}(\phi + \frac{\pi}{2})]^T \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} l_d \dot{\phi}$$

$$\left(\vec{r}_{d/w}|_A \right)^\cdot\cdot = \left(\vec{r}_{p/w}|_A \right)^\cdot\cdot + \left(\vec{r}_{d/p}|_A \right)^\cdot\cdot = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \ddot{x} + [\mathcal{O}(\phi + \pi)]^T \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} l_d \dot{\phi}^2 + [\mathcal{O}(\phi + \frac{\pi}{2})]^T \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} l_d \ddot{\phi}$$

$$\left(\vec{r}_{c/\sigma_2}|_A \right)^\cdot = [\mathcal{O}(\phi + \frac{\pi}{2})]^T \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} l_c \dot{\phi}$$

$$\left(\ddot{\vec{r}}_{c/\sigma_2}|_A\right) = \left[\mathcal{O}\left(\phi + \frac{\pi}{2}\right)\right]^T \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} l_c \ddot{\phi} + [\mathcal{O}(\phi + \pi)]^T \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} l_c \dot{\phi}^2$$

$$T_d = \frac{1}{2} m \left(\dot{\vec{r}}_{d/w}|_A\right)^T \left(\dot{\vec{r}}_{d/w}|_A\right), \quad \left(\dot{\vec{r}}_{d/w}|_A\right) = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \dot{x} + \left[\mathcal{O}\left(\phi + \frac{\pi}{2}\right)\right]^T \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} l_d \dot{\phi}$$

$$\frac{\partial T_d}{\partial \dot{x}} = \frac{1}{2} m \left(\frac{\partial}{\partial \dot{x}} \left(\dot{\vec{r}}_{d/w}|_A\right)\right)^T \left(\dot{\vec{r}}_{d/w}|_A\right) + \frac{1}{2} m \left(\dot{\vec{r}}_{d/w}|_A\right)^T \frac{\partial}{\partial \dot{x}} \left(\dot{\vec{r}}_{d/w}|_A\right)$$

$$= \frac{1}{2} m \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}^T \left(\dot{\vec{r}}_{d/w}|_A\right) + \frac{1}{2} m \left(\dot{\vec{r}}_{d/w}|_A\right)^T \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

$$= m \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}^T \left(\dot{\vec{r}}_{d/w}|_A\right)$$

$$\frac{\partial}{\partial t} \frac{\partial T_d}{\partial \dot{x}} = m \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}^T \left(\ddot{\vec{r}}_{d/w}|_A\right)$$

$$= m \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}^T \left(\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \ddot{x} + [\mathcal{O}(\phi + \pi)]^T \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} l_d \dot{\phi}^2 + \left[\mathcal{O}\left(\phi + \frac{\pi}{2}\right)\right]^T \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} l_d \ddot{\phi} \right)$$

$$\frac{\partial T_d}{\partial x} = 0$$

$$\left(\dot{\vec{r}}_{c/w}|_A\right) = \left(\dot{\vec{r}}_{d/w}|_A\right) - \left(\dot{\vec{r}}_{d/\sigma_2}|_A\right) + \left(\dot{\vec{r}}_{c/\sigma_2}|_A\right)$$

$$\theta_4 = g(\theta_2), \quad \dot{\theta}_4 = \left(\frac{\partial g}{\partial t}(\theta_2)\right) \dot{\theta}_2, \quad \ddot{\theta}_4 = \left(\frac{\partial^2 g}{\partial t^2}(\theta_2)\right) \dot{\theta}_2^2 + \left(\frac{\partial g}{\partial t}(\theta_2)\right) \ddot{\theta}_2$$

Code

Exercise One: Inclined Bar

```
clear;clc;close all;

tspan=[0 10];
q0=[0;0;pi/2-1e-2;0];
% q0=[0;1;pi/2;0];
options = odeset('RelTol',1e-4, ...
    "AbsTol",1e-4, ...
    "Events",@Ver2EventsFcn);
[t,q]=ode45(@Ode45InclinedBarVer2,tspan
,q0,options);

n=size(q,1);
N=zeros(n,1);
f=zeros(n,1);
s=zeros(n,1);
for i=1:n

[~,N(i),f(i),s(i)]=InclinedBarVer2(t(i)
,q(i,:));
end

figure;
set(gcf,'visible','on');

subplot(5,1,1);
plot(t,q(:,1),'-o');
ylabel("x [m]");

subplot(5,1,2);
plot(t,rad2deg(q(:,3)),'-o');
ylabel("theta [deg]");

subplot(5,1,3);
plot(t,N,'-o');
ylabel("N [N]");

subplot(5,1,4);
plot(t,f,'-o');
ylabel("f [N]");

subplot(5,1,5);
plot(t,s,'-o');
hold on;
yline(0);
hold on;
yline(1);
hold on;
yline(-1);
```

```
hold on;
ylabel("s [ ]");
xlabel("t [s]")
```

Superpose Graph

```
l=1; %m
figure;
set(gcf,'Visible','on');
x=q(:,1);
theta=q(:,3);

n_frame=10;
frameIndexArray=ones(1,n_frame);
tIdeal=linspace(min(t),max(t),n_frame);
for i=1:n_frame
    [~,frameIndexArray(i)]=min(abs(t-
tIdeal(i)));
end
for i=frameIndexArray

line([x(i),x(i)+cos(theta(i))*l], ...
    [0,sin(theta(i))*l]);
    hold on;
%
plot((x(i)+x(i)+cos(theta(i))*l)/2,...
%     sin(theta(i))*l/2,'ok')
end
axis equal
xlabel("i_A Direction");
ylabel("-k_A Direction");
title("mu=0.1")
```

Function InclineBarVer2

```
function [q_dot,N,f,s] =
InclineBarVer2(t,q)
l=1; %m
m=1; %kg
g=9.8; %N/kg
mu=0.1; %N/N
% mu=0;
J=1/12*m*l^2; %kgm2
epsilon=1e-2; %m/s

x=q(1);
x_dot=q(2);
theta=q(3);
theta_dot=q(4);

if abs(x_dot)>epsilon
    s=sign(x_dot);
```

```

else
    s=x_dot/epsilon;
%    s=0;
end

A=[0,    0.5*1*cos(theta),    -1/m, 0;
    1,  -0.5*1*sin(theta),  0,  -1/m;
    0,  J,  0.5*1*cos(theta),  -
0.5*1*sin(theta);
    0,  0,  s*mu,  1];
b=[theta_dot^2*0.5*1*sin(theta)-g;
    theta_dot^2*0.5*1*cos(theta);
    0;
    0];
c=linsolve(A,b);

N=c(3);
f=c(4);
q_dot=[q(2);c(1);q(4);c(2)];

```

```
end
```

Function Ode45InclineBarVer2

```

function [q_dot] =
Ode45InclinedBarBer2(t,q)
[q_dot,~,~,~] = InclinedBarVer2(t,q);
end

```

Function Ode45InclineBarVer2

```

function [theta,isterminal,direction] =
Ver2EventsFcn(t,q)
theta = q(3); % The value that we want
to be zero
isterminal = 1; % Halt integration
direction = 0; % The zero can be
approached from either direction
end

```

Exercise Two: Jumping Rail

```
clear;clc;close all;
x=linspace(-0.1,0.1,1e2);
k=100;

% b_array=[-1e-3,4e-2,0.1];
b_array=-1e-3
a_array=0.5./(b_array.*exp(-
0.1./b_array)+0.1-b_array);

for i=1:length(b_array)
    b=b_array(i)
    a=a_array(i)
    E_p_linear=k*x;
    E_p_exp=a*(exp(x/b)-1);
    E_linear=0.5*k*x.^2;
    E_exp=a*(b*exp(x./b)-x-b);

    figure;
    set(gcf,'visible','on');
    subplot(2,1,1);
    plot(x,E_p_exp,'o');
    hold on;
    subplot(2,1,2);
    plot(x,E_exp,'o');
    hold on;

    subplot(2,1,1);
    plot(x,E_p_linear,'-');
    legend();
    subplot(2,1,2);
    plot(x,E_linear,'-');
    legend();
end
```

Contour Plot Code

```
clear;clc;close all;

N_I=10;
N_J=10;

[theta_0_mesh,mu_mesh]=meshgrid(deg2rad(linspace(30,90,N_I)),linspace(0.3,1,N_J));
v_bar_hori=zeros(N_I,N_J);
v_bar_vert=zeros(N_I,N_J);
```

```
theta_dot_takeoff=zeros(N_I,N_J);

for I=1:N_I
    for J=1:N_J
        "(I,J) = "
        "+num2str(I)+", "+num2str(J)

        para=struct();
        para.g=9.8; %N/kg
        para.M=0.4; %kg
        para.m=0.1; %kg
        % para.mu=0.3; %N/N
        para.mu=mu_mesh(I,J);
        para.l=0.5; %m
        para.J=1/12*1*para.l^2;

        tspan=[0 2];
        q0=[0;0;para.l/2-
0.1;0;theta_0_mesh(I,J);0];
        options = odeset('RelTol',1e-
7, ...
        "AbsTol",1e-7, ...

        "Events",@RailJumpingEventsFcn);

        [t,q]=ode23(@(t,q)OdeRailJumping(t,q,para)
        ,tspan,q0,options);

        n=size(q,1);
        N=zeros(n,1);
        f=zeros(n,1);
        s=zeros(n,1);
        for i=1:n

            [~,N(i),f(i),s(i)]=RailJumping(t(i),q(i,:),
            ,para);

            end

        index_separate=find(N<=0,1,'first');
        q=q(1:index_separate,:);
        t=t(1:index_separate);
        n=size(q,1);
        N=zeros(n,1);
        f=zeros(n,1);
        s=zeros(n,1);
        for i=1:n

            [~,N(i),f(i),s(i)]=RailJumping(t(i),q(i,:),
            ,para);

            end

        % Demux
        x=q(:,1)';
        x_dot=q(:,2)';
        y=q(:,3)';
        y_dot=q(:,4)';
```

```

        theta=q(:,5)';
        theta_dot=q(:,6)';
        % Velocity Calculation
        v_d_w_A=[x_dot+y_dot.*cos(theta)-
y.*sin(theta).*theta_dot;
        zeros(1,n);
        -y_dot.*sin(theta)-
y.*cos(theta).*theta_dot];
        v_c_w_A=[x_dot-
0.5*para.l*sin(theta).*theta_dot;
        zeros(1,n);
        -
0.5*para.l*cos(theta).*theta_dot];

v_center_w_A=(para.M*v_d_w_A+para.m*v_c_w_
A)./(para.M+para.m);
        % figure;
        % set(gcf,'visible','on');
        % plot(t,v_center_w_A(1,:),'-o');
        % hold on;
        % plot(t,v_center_w_A(3,:),'-o');
        % hold on;

v_bar_hori(I,J)=v_center_w_A(1,end);
        v_bar_vert(I,J)=-
v_center_w_A(3,end);

theta_dot_takeoff(I,J)=theta_dot(end);
        end
end

```

```

figure("position",[0,0,800,300]);
set(gcf,'visible','on');
titleText=["v_{bar,hori} [m/s]";
        "v_{bar,vert} [m/s]";
        "theta_{dot} [deg/s]"];
for i=1:3
    switch i
        case 1
            Z=v_bar_hori;
        case 2
            Z=v_bar_vert;
        case 3
            Z=rad2deg(theta_dot_takeoff);
    end
    subplot(1,3,i);

contourf(rad2deg(theta_0_mesh),mu_mesh,Z,
...
        'ShowText','on');
    hold on;
    xlabel("theta_0 [deg]");
    ylabel("mu [ ]");
    title(titleText(i));

xlim(rad2deg([min(theta_0_mesh,[],'all'),m
ax(theta_0_mesh,[],'all')]));
    if i==1

```

```

        mu_temp=linspace(0.3,1.0,1e2);
        theta0_temp=atan(1./mu_temp);

plot(rad2deg(theta0_temp),mu_temp,'r--
','LineWidth',2);
        hold on;
        plot(-
10+rad2deg(theta0_temp),mu_temp,'b--
','LineWidth',2);
        hold on;
        end
end

```

One Case Detailed Plot

```

figure;
set(gcf,'visible','on');

subplot(3,3,1);
plot(t,q(:,1),'-o');
ylabel("x [m]");

subplot(3,3,4);
plot(t,q(:,3),'-o');
ylabel("y [m]");

subplot(3,3,7);
plot(t,rad2deg(q(:,5)),'-o');
ylabel("theta [deg]");

subplot(3,3,2);
plot(t,q(:,2),'-o');
ylabel("x_dot [m/s]");

subplot(3,3,5);
plot(t,q(:,4),'-o');
ylabel("y_dot [m/s]");

subplot(3,3,8);
plot(t,rad2deg(q(:,6)),'-o');
ylabel("theta_dot [deg/s]");

subplot(3,3,3);
plot(t,N,'-o');
ylabel("N [N]");

```

```

subplot(3,3,6);
plot(t,f,'-o');
ylabel("f [N]");

subplot(3,3,9);
plot(t,s,'-o');
hold on;
yline(0);
hold on;
yline(1);
hold on;
yline(-1);
hold on;
ylabel("s [ ]");
xlabel("t [s]");

```

One Case Superposition Plot

```

l=para.1; %m
figure;
set(gcf,'Visible','on');
x=q(:,1);
y=q(:,3);
theta=q(:,5);

n_frame=4;
frameIndexArray=ones(1,n_frame);
tIdeal=linspace(min(t),max(t),n_frame);
for i=1:n_frame
    [~,frameIndexArray(i)]=min(abs(t-
tIdeal(i)));
end
j=1;
for i=frameIndexArray
    subplot(4,1,j);
    line([x(i),x(i)+cos(theta(i))*1], ...
        [0,sin(theta(i))*1], ...
        'color','b');
    hold on;
    plot(x(i)+cos(theta(i))*y(i),...
        sin(theta(i))*y(i),...
        'ok');
    hold on;

    line([x(i)+cos(theta(i))*y(i),x(i)+cos(theta(i))*1/2], ...
        [sin(theta(i))*y(i),sin(theta(i))*1/2], ...
        'color','r');
    hold on;
    % xlabel("i_A Direction");
    % ylabel("-k_A Direction");
    xlim([-0.5 0.5])
    ylim([0 0.3])

```

```

axis equal
xlabel("t = "+num2str(t(i))+ "s");
j=j+1;
end

```

Two Model Comparison Study

```

clear;clc;close all;

tspan=[0 1];
q0_jump=[0;0;0.5-9.8/1e5;0;pi/2-1e-1;0;];
options = odeset('RelTol',1e-6, ...
    "AbsTol",1e-6, ...
    "Events",@RailJumpingEventsFcn);
[t_jump,q_jump]=ode23(@OdeRailJumping,tspan,q0_jump,options);

q0_inclined=[0;0;pi/2-1e-1;0;];
% q0=[0;1;pi/2;0;];
options = odeset('RelTol',1e-4, ...
    "AbsTol",1e-4, ...
    "Events",@Ver2EventsFcn);
[t_inclined,q_inclined]=ode23(@Ode45InclinedBarVer2,tspan,q0_inclined,options);

figure;
set(gcf,'visible','on');

subplot(2,1,1);
plot(t_jump,q_jump(:,1),'o');
hold on;
plot(t_inclined,q_inclined(:,1));
hold on;
xlabel("t [s]");
ylabel("x [m]");
legend(["Jumping Rail";"Inclined Bar"]);

subplot(2,1,2);
plot(t_jump,rad2deg(q_jump(:,5)),'o');
hold on;
plot(t_inclined,rad2deg(q_inclined(:,3)));
hold on;
xlabel("t [s]");
ylabel("theta [deg]");
legend(["Jumping Rail";"Inclined Bar"]);

```

function
[q_dot,N,f,s]=RailJumping(t,q)

```
function q_dot=OdeRailJumping(t,q,para)
[q_dot,~,~,~]=RailJumping(t,q,para);
end
```

function [q_dot,N,f,s]=RailJumping(t,q)

```
function
[q_dot,N,f,s]=RailJumping(t,q,para)
% Demux
x=q(1);
x_dot=q(2);
y=q(3);
y_dot=q(4);
theta=q(5);
theta_dot=q(6);
% Constants
g=para.g;
M=para.M;
m=para.m;
mu=para.mu;
l=para.l;
J=para.J;
% Variables
Epy=funEpy(y,l/2);
epsilon=1e-2; %m/s
if abs(x_dot)>epsilon
    s=sign(x_dot);
else
    s=x_dot/epsilon;
end
% A and b
A=[M+m,M*cos(theta),-M*y*sin(theta)-
0.5*m*l*sin(theta),0,-1;
    M*cos(theta),M,0,0,0;
    -M*y*sin(theta)-
0.5*m*l*sin(theta),0,M*y^2+0.25*m*l^2+J,0,
0;
    0,M*sin(theta),0.5*m*l*cos(theta)+M*y*cos(
theta),-1,0;
    0,0,0,s*mu,1];
b=[y_dot*theta_dot*(M*sin(theta))+theta_dot
t^2*(M*y*cos(theta)+0.5*m*l*cos(theta));
    theta_dot^2*M*y-M*g*sin(theta)-Epy;
```

```
    -y_dot*theta_dot*2*M*y-
0.5*m*g*l*cos(theta)-M*g*y*cos(theta);
    -g*(M+m)-
y_dot*theta_dot*(2*M*cos(theta))+theta_dot
^2*(M*y*sin(theta)+0.5*m*l*sin(theta));
    0];
chi=linsolve(A,b);
% q_dot
q_dot=zeros(6,1);
q_dot(1)=x_dot;
q_dot(2)=chi(1);
q_dot(3)=y_dot;
q_dot(4)=chi(2);
q_dot(5)=theta_dot;
q_dot(6)=chi(3);
N=chi(4);
f=chi(5);
end
```

function Epy=funEpy(y,y0)

```
function Epy=funEpy(y,y0)
k=1e2;
Epy=-k*(y0-y);

% b=[-1e-1];
% a=0.5./(b.*exp(-0.1./b)+0.1-b);
% Epy=a*(exp((y-y0)/b)-1);
end
```

function [theta,isterminal,direction] = RailJumpingEventsFcn(t,q)

```
function [theta,isterminal,direction] =
RailJumpingEventsFcn(t,q)
    theta = q(5); % The value that we want
to be zero
    isterminal = 1; % Halt integration
    direction = 0; % The zero can be
approached from either direction
end
```

Exercise 3: Jacobian Generator

```
clear;clc;close all;
n=1e2;
theta2=deg2rad(linspace(30,-30,n));

figure;
set(gcf,'Visible','on');

pm=1;
a=1;
b=0.3;
c=1;
d=2;
Ax=a*cos(theta2);
Ay=a*sin(theta2);
S=(a^2-b^2+c^2-d^2)./(2*(Ax-d));
C=Ay.^2./(Ax-d).^2+1;
Q=2*Ay.*(d-S)./(Ax-d);
R=(d-S).^2-c^2;
By=(-Q+pm.*sqrt(Q.^2-4.*C.*R))./(2*C);
Bx=S-By.*Ay./(Ax-d);
% theta3=atan2((By-Ay),(Bx-Ax));
% theta4=atan2(By,(Bx-d));
theta3=atan((By-Ay)./(Bx-Ax));
theta4=atan(By./(Bx-d));
```

```
index=(imag(By)==0);
```

```
subplot(2,1,1);
plot(rad2deg(theta2(index)),rad2deg(theta3(index)));
hold on;
subplot(2,1,2);
plot(rad2deg(theta2(index)),rad2deg(theta4(index)));
hold on;
```

Kinematics Plotter

```
figure;
set(gcf,'visible','on');
subplot(2,1,1);
plot(rad2deg(theta2(index)),rad2deg(theta3(index)),'-o');
subplot(2,1,2);
plot(rad2deg(theta2(index)),rad2deg(theta4(index)),'-o');
```

```
O2=zeros(2,n);
O4=[ones(1,n)*d;zeros(1,n)];
A=[Ax;Ay];
B=[cos(theta4)*c+d;sin(theta4)*c];
B=[Bx;By];
C=(A+B)/2;
e=0.5;
O2_E=[zeros(1,n);ones(1,n)*e];
A_E=[Ax;Ay+e];
C_E=[C(1,:);C(2,:)+e];
P=2.5*C-1.5*C_E;
```

```
N=4;
```

```
figure("Position",[0,0,1600,300]);
set(gcf,'visible','on');
index_frame=floor(linspace(find(index==1,1,'first'),find(index==1,1,'last'),N));
j=1;
for i=index_frame
    subplot(1,N,j);

    line([O2(1,i),A(1,i)],[O2(2,i),A(2,i)] ...
        , 'color','k','linewidth',2);
    hold on;

    line([A(1,i),B(1,i)],[A(2,i),B(2,i)] ...
        , 'color','k','linewidth',2);
    hold on;

    line([B(1,i),O4(1,i)],[B(2,i),O4(2,i)] ...
        , 'color','k','linewidth',2);
    hold on;

    line([O4(1,i),O2(1,i)],[O4(2,i),O2(2,i)] ...
        , 'color','g','linewidth',2);
    hold on;
    plot(C(1,i),C(2,i),'ok');
    hold on;
```

```
line([O2(1,i),O2_E(1,i)],[O2(2,i),O2_E(2,i)] ...
    , 'color','g','linewidth',2);
    hold on;

    line([A_E(1,i),O2_E(1,i)],[A_E(2,i),O2_E(2,i)] ...
        , 'color','b','linewidth',2);
```

```

    hold on;

line([A_E(1,i),C_E(1,i)],[A_E(2,i),C_E(2,i)
]) ...
    , 'color','b', 'linewidth',2);
    hold on;

line([A_E(1,i),A(1,i)],[A_E(2,i),A(2,i)] .
..
    , 'color','b', 'linewidth',2);
    hold on;
    plot(C_E(1,i),C_E(2,i), 'ok');
    hold on;

```

```

line([C_E(1,i),P(1,i)],[C_E(2,i),P(2,i)] .
..
    , 'color','r', 'linewidth',2);

    j=j+1;
    xlim([-0.5 2.5]);
    ylim([-1,1.5]);
end

```